Single-Image Multiplicative Layer Separation for Physical Material Estimation of Stained and Art Glass: An Exhaustive Review

The challenge of extracting per-pixel physical material maps—specifically RGB transmittance, scattering strength, and surface relief—from a single uncontrolled photograph of stained or art glass represents a profoundly ill-posed inverse rendering problem. Unlike traditional opaque object reconstruction, where the observed radiance is a function of incident illumination and surface reflectance, the appearance of translucent and transparent media is overwhelmingly dominated by transmission. The image formation model is approximately multiplicative: the observed image $I(x)$ is the product of the spatially varying transmittance of the glass $T(x)$ and the radiance of the background scene $B(x)$ visible through it. When subsurface scattering $\sigma_s(x)$ is introduced, the background is additionally subjected to a spatially varying low-pass filter, further entangling the background radiance with the material properties.

This report provides an exhaustive, multi-disciplinary survey of the literature from 2022 to 2026. The analysis is specifically oriented around the multiplicative layer separation problem and the extraction of capture-invariant material representations suitable for physically-based rendering (PBR) engines. The review spans generative priors, dual-stream reflection removal, appearance capture, synthetic-to-real domain adaptation, multi-illumination consistency, test-time optimization, uncertainty estimation, and surface relief recognition.

1. Intrinsic Image Decomposition and Single-Image Inverse Rendering

The domain of intrinsic image decomposition (IID) aims to factor an image into its fundamental photometric components, typically albedo and shading. Historically constrained by optimization frameworks reliant on Retinex theory, the field has been revolutionized by the integration of large-scale diffusion models. These foundation models encapsulate profound priors regarding natural image statistics, geometry, and lighting. However, the standard IID formulation assumes an additive or purely Lambertian opaque world, fundamentally diverging from the multiplicative transmission physics of stained glass.

The *Marigold* framework[1] fine-tunes latent diffusion models like Stable Diffusion for dense prediction tasks, including depth, surface normals, and intrinsic image decomposition. While it sets a contemporary standard for high-resolution dense mapping of albedo and diffuse shading, its foundational reliance on additive, Lambertian assumptions makes it highly inapplicable to the multiplicative separation of transmissive backgrounds, as the model

inevitably interprets high-frequency background features as foreground surface albedo. Code and model weights for Marigold are publicly available.
Addressing the inherent ambiguities between lighting and material properties, the *Intrinsic Image Diffusion* architecture[5] utilizes a conditional generative model to sample multiple probabilistic material explanations (albedo, roughness, metallic) from a single indoor scene image. Although its probabilistic exploration of the solution space is conceptually valuable for the deeply ill-posed stained glass problem, its specific generative priors are trained exclusively on opaque indoor scenes, rendering it ineffective for modeling transmitted light. Code and weights for this method are publicly available.
Further leveraging foundational priors, *IntrinsicDiffusion*[7] introduces a ControlNet-based architecture built on a pre-trained text-to-image generation model to jointly predict shading, albedo, and surface normals. This collaborative prediction improves the overall consistency of opaque material maps, yet the complete absence of a transmission or opacity channel prevents it from solving the $I(x) = T(x) \cdot B(x)$ problem. Code and weights are publicly available. Similarly, the *IntrinsicAnything* approach[9] learns diffusion priors specifically for albedo and specular shading to regularize inverse rendering under unknown illumination. While it excels at decoupling environment lighting from surface reflectance via a coarse-to-fine multi-view consistent strategy, it does not model refractive or transmissive light transport, making it unsuitable for separating backgrounds from stained glass transmittance. Code and weights are publicly available.
Bridging the gap between decomposition and synthesis, the *RGB↔X* framework[12] enables bidirectional mapping between RGB images and intrinsic PBR channels by exploring the connections between diffusion models and realistic rendering. The capacity of this network to hallucinate missing appearance properties is highly relevant for filling in ambiguous transmittance data, but the framework is currently constrained to opaque bidirectional reflectance distribution functions (BRDFs) and does not separate multiplicative background layers. Code and weights are publicly available.
Finally, *Ouroboros*[15] introduces a framework composed of two single-step diffusion models that handle forward and inverse rendering with mutual reinforcement via cycle-consistency. This cycle-consistent training mechanism is theoretically highly applicable to the multiplicative separation problem, as it could enforce that the separated glass and background maps re-render perfectly into the input photo without requiring ground-truth paired data, though its current implementation is restricted to opaque parameters. Code and weights are publicly available. *Uni-Renderer*[22] utilizes a similar dual-stream diffusion framework and cycle-consistent constraints to reinforce the correlation between intrinsic properties and rendered images. While effective for opaque decomposition, its lack of transmission handling limits direct application to glass. Code availability is unknown.

2. Reflection Removal and Image Layer Separation

Reflection removal is the mathematical inverse and structural sibling of the multiplicative transmissive problem. While reflection removal typically models the image formation process as an additive mixture of a transmission layer and a reflection layer ($I = T + R$), the stained glass problem is multiplicative ($I = T \cdot$). Crucially, in logarithmic space, the multiplicative problem becomes additive ($\log I = \log T + \log B$). Therefore, the advanced dual-stream architectures and diffusion guidance mechanisms developed for reflection removal offer the most promising architectural templates for multiplicative layer separation.

The *Flash-Split* framework[23] separates transmitted and reflected light using a dual-branch diffusion model operating in a low-dimensional latent space, conditioned on misaligned flash and no-flash image pairs. If modified to accept a single image (or an optional context image) and mathematically transformed into log-space, its recursive latent separation mechanism could highly effectively separate high-frequency background details from the low-frequency glass transmittance. Code and weights are publicly available.

Focusing on foundation model adaptation, *WindowSeat*[27] repurposes a massive diffusion-transformer (DiT) via lightweight LoRA adapters to predict clean transmission layers from reflection-contaminated inputs in a single step. The use of a physically based rendering (PBR) pipeline simulating index of refraction, thickness, and subsurface scattering to generate synthetic training data makes its methodology directly and highly transferable to training models for art glass separation. Code and weights are publicly available.

Addressing the critical role of color spaces, the system described in *Removing Reflections from RAW Photos*[34] operates strictly on linear RAW photos, utilizing an optional contextual photo to disambiguate layers, and is trained exclusively on physically accurate synthetic mixtures incorporating Fresnel attenuation and defocus blur. Its insistence on linear, scene-referred color spaces is absolutely critical for the multiplicative $T \cdot$ problem, as operating on gamma-compressed or tone-mapped 8-bit images mathematically destroys the true multiplicative relationship between light fields. Code is not publicly available, as it is implemented in commercial Adobe software.

To address color bias and ghosting introduced by colored glass, the *Polarization State Tracing* model[39] traces polarized light along multiple propagation paths and crucially accounts for wavelength-selective absorption within the medium. The explicit modeling of absorption (obeying the Beer-Lambert law) makes it highly relevant for estimating the true physical transmittance $T(x)$ of stained glass, provided that polarization sensor data is available at capture. Code availability is standard for CVPR publications but not explicitly verified.

Exploring purely mathematical separation, *Fast and Accurate Multiplicative Decomposition* by Soncco et al.[43] presents a variational model for multiplicative image decomposition to remove interference fringes, utilizing smoothed total variation operators and one-dimensional Fourier transforms. While mathematically formulated precisely for multiplicative separation, its reliance on specific geometric priors (horizontal, periodic fringes) prevents its application to the unstructured, high-frequency natural backgrounds seen behind art glass. Code availability is unknown.

Feature	Flash-Split	WindowSeat	RAW Photos	Pol. State Tracing	Mult. Decomp
Mixing Model	Additive Latent	Additive Latent	Linear RAW (Additive)	Polarization Physics	Multiplicative
Inputs Required	Flash/No-Flash Pairs	Single Image	RAW + Context Image	Polarization Data	Single Image
Primary Prior	Dual-Branch Diffusion	Pre-trained DiT (LoRA)	CNN / Synthetic priors	Wavelength Absorption	Total Variation / Fourier
Applicability to T.	High (Log-space adaptation)	High (PBR training pipeline)	Critical (Linear color space)	High (Beer-Lambert focus)	Low (Geometric restriction)

3. Appearance Capture of Translucent and Transparent Materials

Capturing the physical properties of translucent materials requires estimating not just surface reflectance, but volumetric scattering and absorption. In the context of art glass, scattering ($\sigma_s(x)$) dictates the degree to which the background is blurred or diffused—such as the hazy subsurface scatter seen in opalescent glass. Single-image estimation of these parameters is historically intractable, prompting the field to rely on multi-view setups, complex laboratory acquisition, or diffusion-based generative augmentation.

The *DIAMOND-SSS* architecture[47] provides a framework for high-fidelity translucent reconstruction that utilizes a diffusion-based data augmentation strategy to synthesize missing viewpoints and relighting conditions from extremely sparse captures. While designed for 3D Gaussian Splatting rather than single-image 2D decomposition, its successful use of geometry-conditioned diffusion models to hallucinate subsurface scattering (SSS) effects demonstrates how generative priors can solve under-constrained volumetric scattering problems. Code availability is unknown.

Addressing flatbed material capture, the method in *Single-image Reflectance and Transmittance Estimation from Any Flatbed Scanner*[48] estimates opacity and transmittance maps alongside surface normals and roughness from a single scanner image using neural intrinsic decomposition principles. While it directly addresses transmittance estimation and models light scattered through the material as a linear attenuation, it assumes uniform, controlled scanner illumination against a flat background, making it inapplicable to uncontrolled phone photos with complex spatial backgrounds. Code availability is unknown.

Operating in 3D space, *TransparentGS*[54] introduces a fast inverse rendering pipeline for transparent objects using transparent Gaussian primitives to handle specular refraction and Gaussian light field probes to encode ambient light and nearby content. It effectively models the complex distortion of backgrounds seen through clear and semi-transparent glass but strictly requires multi-view input and environment matting, rendering it unsuitable for the single-image formulation. Code and weights are publicly available.

Targeting depth rather than appearance, the *SeeClear* framework[60] converts transparent objects into generative opaque images using a diffusion-based opacification module, subsequently feeding these geometry-consistent shapes into off-the-shelf monocular depth estimators. While targeted at depth estimation, the concept of algorithmically "opacifying" a transparent surface by replacing refractive appearances with diffuse textures is a highly relevant analogue to isolating the material $T(x)$ layer from the background. Code and weights are publicly available.

From a physical optics perspective, the *Polarized Inverse Scattering for SSS Estimation* method[65] uses polarization cues to jointly estimate shape and subsurface scattering parameters, leveraging the physical insight that SSS affects the ratio between diffuse and specular polarization due to multiple scattering events depolarizing the light. The reliance on a specialized polarization camera eliminates it from the standard smartphone photo requirement, though the physical insight regarding how scattering diffuses background information is valuable. Code availability is unknown.

Finally, foundational work in *VITRAIL: Acquisition, Modelling and Rendering of Stained Glass*[66] establishes that light transport in historical stained glass is highly heterogeneous due to bubbles, corrosion, and grisaille paint, requiring the acquisition of material-dependent dictionaries and light transport matrices. This underscores the complexity of stained glass microgeometry, but their reliance on laboratory-controlled multi-illumination setups prevents its use for in-the-wild capture. Code availability is unknown.

4. Synthetic-to-Real Training for Material Estimation

The acquisition of ground-truth paired data for layer separation in the wild is physically impossible. One cannot capture a perfectly aligned image of stained glass with a background, the glass perfectly isolated, and the background perfectly unobstructed under identical lighting conditions without pixel shifts. Consequently, algorithmic progress relies entirely on rendering-in-the-loop, analysis-by-synthesis, and cycle-consistency supervision to cross the domain gap.

The PBR pipeline developed for *WindowSeat*[27] demonstrates the power of synthetic data generation. To train their diffusion transformer for reflection removal, the authors constructed a scalable physically based rendering pipeline in Blender using the Principled BSDF to simulate realistic glass reflections, volumetric scattering, and ghosting artifacts over natural scenes. This methodology is directly and immediately applicable to the stained glass problem: a parallel PBR pipeline could generate millions of synthetic training pairs of I, T, and B by rendering procedural, texture-mapped glass planes against diverse HDRI environments. Code and data generation scripts are publicly available.

The necessity of precise simulation is reinforced by *Removing Reflections from RAW Photos*[34], which relies entirely on synthetic mixtures of real RAW images, explicitly simulating Fresnel attenuation, double reflections, and physically calibrated defocus blur. The authors validate that training exclusively on high-fidelity, linear RAW synthetic data outperforms architectural variations among prior works, confirming that a synthetic-to-real domain strategy is viable for multiplicative layer separation only if the physical mixing is modeled accurately. Code is not publicly available.

To bypass the need for ground-truth paired data entirely, the *Ouroboros* framework[15] employs a cycle-consistency mechanism between an inverse rendering model (decomposition) and a forward rendering model (synthesis). Applying this render-in-the-loop cycle-consistency to stained glass would require predicting $T(x)$, $\sigma_s(x)$, and $B(x)$ from a real photo, rendering them together via a differentiable multiplicative and scattering operation, and minimizing the reconstruction loss against the original input photo, thereby allowing self-supervised learning on unannotated real-world data. Code and weights are publicly available.

Highlighting the scale required for synthetic-to-real transfer, the *SeeClear-396k Dataset*[60] was constructed to train a generative opacification module, comprising 396k paired transparent-opaque renderings with aligned depth, normals, and object masks. This massive dataset exemplifies the necessity of scale to teach diffusion models the complex, non-Lambertian physics of refraction and transmission, strongly supporting the requirement for a similarly massive synthetic dataset for transmissive art glass materials. The dataset is publicly available. Furthermore, research in *Synthetic Active Learning for Parts Recognition*[70] proposes a closed-loop framework using domain randomization to adaptively refine synthetic data generation based on model weaknesses without real samples, which could be used to iteratively improve procedural glass texture generation. Code availability is unknown.

5. Lighting and Capture Invariance Objectives

A core requirement of physical material estimation is capture invariance: the estimated material maps (T and σ_s) must remain consistent regardless of the incident illumination or the specific background placed behind the glass. Recent literature highlights the critical use of multi-illumination datasets and spatial harmonization to enforce this consistency.

The *LightCity* dataset[71] provides synthetic urban scenes with diverse, highly controllable illumination conditions and complex indirect lighting, specifically benchmarked to test the intrinsic coherency of scene decomposition models across varying lighting. The benchmarking proves that current models heavily lack multi-illumination consistency, highlighting the absolute necessity for specialized consistency losses during training to ensure illumination variations are

not baked into the physical material properties. The dataset is publicly available.
Addressing indoor multi-illumination, *LuxRemix*[73] presents a generative image-based light decomposition model that factorizes complex indoor illumination into constituent light sources, utilizing multi-view harmonization to propagate the decomposition consistently. While focused on multi-view relighting, the objective of independent manipulation of lighting components perfectly aligns with the goal of separating the environment illumination baked into the background layer $B(x)$ from the intrinsic color of the glass $T(x)$. Code availability is unknown.
Representing the apex of controlled real-world data, *OLATverse*[75] is a massive dataset of 9 million images covering 765 objects under precisely controlled One-Light-at-a-Time (OLAT) conditions. While primarily focused on opaque and mildly reflective objects, it establishes the gold standard for evaluating capture-invariant material estimation, demonstrating the rigorous empirical validation required to prove true physical relightability. The dataset is publicly available. Similarly, *OpenIllumination*[78] provides a multi-illumination dataset for real objects captured under diverse lighting, acting as a crucial data structure for enforcing consistency losses during network training (i.e., penalizing the model if the predicted $T(x)$ fluctuates across different illuminations of the same object). The dataset is publicly available.
Exploring the intersection of motion and illumination, *LumiMotion*[80] demonstrates that dynamics—specifically rigidly moving objects—provide stronger cues for disentangling material from illumination than static views, improving albedo estimation significantly. This suggests that multi-view or video priors heavily aid disentanglement, which is highly relevant if the input phone photos can be extracted from brief video bursts or sweeps. Code availability is unknown. Furthermore, *LightCrafter*[81] reformulates video relighting as a translation of PBR proxy videos, baking illumination targets into the proxy. This is relevant for maintaining temporal consistency when relighting the extracted physical material maps in a PBR engine. Code availability is unknown.

6. Test-Time Optimization with Generative Priors

Because feed-forward layer separation networks often fail on out-of-distribution real-world images due to the extreme domain gap, test-time optimization (TTO) leverages pre-trained generative models as powerful structural priors to solve the inverse problem iteratively at inference time. Diffusion Posterior Sampling (DPS) has emerged as the leading technique in this domain.
The *Spectral Diffusion Posterior Sampling* framework[82] integrates a learned diffusion prior with a physics-based forward model to perform one-step material decomposition for dual-layer CT systems under sparse-view conditions. By alternating between reverse diffusion sampling and iterative optimization to ensure measurement consistency, this logic is highly applicable to stained glass: the model can be forced to obey the physical constraint that the product of the generated $T(x)$ and $B(x)$ maps must equal the observed image $I(x)$ at each diffusion time step. Code availability is unknown.

To accelerate this process, *Anchored Discrete Posterior Sampling*[84] introduces a discrete diffusion posterior sampler that provides faster inference and training-free guidance compared to continuous diffusion, overcoming issues associated with split Gibbs samplers. For the multiplicative layer separation problem, discrete posterior sampling could be highly effective for hallucinating the high-frequency, discrete structural elements of the background $B(x)$ while maintaining the smooth, low-frequency color variations of the transmittance map $T(x)$. Code and weights are publicly available.

Addressing the complexities of real-world optics, recent advancements in *General Diffusion Posterior Sampling for Noisy Inverse Problems*[85] extend DPS to handle nonlinear inverse problems without strict measurement consistency projections, allowing for much better handling of noisy or complex conditions. This is crucially applicable to the stained glass problem because background blurring resulting from subsurface scattering (σ_s) creates a highly nonlinear image formation model, meaning a strict linear projection would fail and relaxed, manifold-constrained gradients are required. Code availability is tied to general open-source DPS repositories.

7. Uncertainty and Confidence Estimation for Dense Regression

When attempting to separate multiplicative layers from a single image, severe mathematical ambiguity is inevitable. For example, if a region of glass is perfectly clear and the background is a flat white wall, it is mathematically impossible to determine if the glass is opaque white and the wall is clear, or vice versa. The network must be capable of quantifying this epistemic uncertainty to prevent confident hallucination of artifacts.

The *Difference Reconstruction Uncertainty Estimation (DRUE)* method[86] addresses the tendency of deep models to be overly confident by measuring uncertainty as the discrepancy between input instances and their reconstructions from auxiliary models or intermediate layers. In the context of layer separation, if the re-multiplied output $T \cdot$ diverges significantly from I, or if multiple generative sampling passes yield widely different T maps, the model can signal high epistemic uncertainty, indicating a fundamentally unresolvable region. Code availability is unknown.

Focusing on spatial variance, *HyperDUM*[87] presents a deterministic uncertainty method utilizing hyperdimensional computing and patch-wise projection to capture spatial uncertainties at the feature level. For transmissive layer separation, patch-wise uncertainty is crucial, as confidence will vary spatially across the image depending on the local presence of background textures, occlusion boundaries, or surface relief on the glass. Code availability is unknown.

To bypass the need for discrete thresholds, the *Harmonious Convergence Estimation* approach[88] calculates intra-batch convergence to estimate confidence in dense regression tasks, utilizing training consistency information. Unlike classification tasks, defining threshold

comparisons for continuous physical variables like transmittance requires specialized algorithms. This approach offers a pathway to ensure that output material maps are accompanied by reliable per-pixel confidence intervals. Code availability is unknown.
Furthermore, research evaluating the *DepthAnything Foundation Model Uncertainty*[89] employs Monte Carlo Dropout, sub-ensembles, and test-time augmentation to extract uncertainty from massive dense regression models. Adapting these established techniques to a diffusion-based separation model would allow the system to output a heatmap denoting where the separation of T and B is ill-posed, deferring to the user or admitting failure rather than producing physically impossible material maps. Code and weights are publicly available.

Uncertainty Method	Core Mechanism	Relevance to Layer Separation	Code Availability
DRUE [cite: 86]	Difference Reconstruction	High (Identifies unresolvable T. regions)	Unknown
HyperDUM [cite: 87]	Patch-wise Hyperdimensional Proj.	High (Maps spatial variance of background texture)	Unknown
Harmonious Conv. [cite: 88]	Intra-batch Convergence	Medium (Handles continuous regression variables)	Unknown
DepthAnything Ens. [cite: 89]	MC Dropout & Sub-ensembles	High (Proven on dense foundation models)	Public

8. Datasets & Benchmarks for Translucent and Reflective Materials

The lack of specialized data is the primary bottleneck in inverse rendering of non-Lambertian surfaces. Recent datasets emphasize transparent, reflective, and multi-illumination conditions to bridge this gap.
The *3DReflecNet* dataset[90] is a massive hybrid corpus exceeding 22 TB, containing over 100,000 synthetic instances and 1,000 real-world objects specifically designed to benchmark

3D reconstruction of reflective, transparent, and low-texture surfaces. While focused on multi-view geometry rather than single-image decomposition, the sheer volume of synthetic transparent rendering makes it a critical resource for pre-training single-image feature extractors on the physical behavior of light through non-Lambertian media. The dataset is publicly available.

Providing temporal context, the *TransPhy3D* corpus[94] is a synthetic video dataset of transparent and reflective scenes containing 11,000 sequences rendered with physically based ray tracing (Blender/Cycles). The inclusion of glass, plastic, and metal materials paired with depth and normal maps provides exactly the kind of synthetic supervision required to teach diffusion models the optical rules of transparency and volumetric scattering. The dataset is publicly available.

Highlighting the difficulties of localization, the *Trans2k* dataset[95] is the first dedicated evaluation dataset for tracking transparent objects, consisting of over 2k sequences annotated with segmentation masks and bounding boxes. It underscores the difficulty of handling objects whose appearance is entirely defined by the background, establishing a baseline for isolating the stained glass sheet against arbitrary backgrounds before performing layer separation. The dataset is publicly available.

Demonstrating the failure of traditional sensing, the *TDoF20* dataset[96] comprises RGB-D images of 20 transparent objects, highlighting the catastrophic failure of commercial depth sensors (like RealSense) due to light refraction and reflection on glass surfaces. It reinforces the need to bypass traditional sensor modalities and rely exclusively on learned generative priors for transmissive material analysis. The dataset is publicly available. Finally, the *GlassPol* dataset[39] contains a wide range of real-world colored glass materials for testing under diverse polarization conditions, serving as a useful validation set for glass absorption properties. The dataset is publicly available.

9. Texture and Surface-Relief Recognition

Art glass frequently exhibits complex surface microgeometry—such as hammered, rippled, or seeded textures—which must be captured to render the glass realistically. From a single image, this requires recognizing surface relief from 2D spatial frequencies.

The computational imaging method for *BTF Classification with Discriminative Illumination*[97] classifies Bidirectional Texture Functions using learned illumination patterns and texture filters, noting that texture is an intrinsic feature for materials. For single-image analysis, while controlled illumination is impossible, their finding that BTF projections are highly sensitive to low spatial frequencies provides a theoretical basis for classifying "hammered" versus "rippled" glass by analyzing the frequency spectrum of the separated transmittance map $T(x)$. Code availability is unknown.

Traditional machine vision analyses of *Statistical Approaches to Surface Texture Classification*[98] rely on the premise that grey levels in an image correspond to surface topography—darker pixels indicate valleys, brighter pixels indicate peaks. In the context of stained glass, surface relief causes micro-refractions that manifest as high-frequency luminance variations. A

dedicated Convolutional Neural Network (CNN) could be trained to classify these luminance variations into standardized glass texture categories. Code availability is unknown.
The *Matador Dataset for Hierarchical Material Recognition*[99] highlights the extreme limitations of existing BRDF/BTF datasets in terms of instance diversity and scale. This research is relevant for ensuring that any texture classification network intended to categorize glass surface finishes trains on a sufficiently diverse taxonomy of surfaces to prevent overfitting to a narrow band of synthetic textures. The dataset is publicly available. Finally, research into *Point Cloud and Volumetric Feature Recognition*[100] details representation strategies for geometric feature recognition. If the surface relief of the glass is estimated as a continuous displacement map, these 3D feature recognition methods could be applied to categorize the physical topology computationally. Code availability is unknown.

Taxonomy of Relevant Literature

The following table categorizes the most relevant works regarding their applicability to the multiplicative transmissive separation problem.

Problem Domain	Paper / Method	Method Family	Supervision / Data	Code	Direct Relevance to T·B Separation
Reflection Removal	Flash-Split (CVPR '25)[23]	Dual-branch Diffusion	Synthetic flash/no-flash	Yes	**High**. Latent separation architectures map exceptionally well to log-space layer separation.
Reflection Removal	WindowSeat (NTIRE '26)[31]	Foundation DiT LoRA	Synthetic PBR (Blender)	Yes	**High**. PBR pipeline design is ideal for generating synthetic $T \cdot$ training

					data.
Reflection Removal	RAW Photos (CVPR '25)[35]	Feed-forward CNN	Synthetic RAW mixtures	No	**Critical**. Proves linear, scene-referred mixing is mathematically necessary for accurate separation.
Reflection Removal	Polarization State Tracing (CVPR '26)[39]	Polarized Transformer	GlassPol Dataset	Unclear	**High**. Explicitly models wavelength absorption and Beer-Lambert transmission in colored glass.
Layer Separation	Multiplicative Decomp (Soncco)[43]	Variational / Fourier	Interferometric images	Unclear	**Medium**. Mathematically aligned (multiplicative), but geometrically restricted to periodic fringes.
Intrinsic Decomp.	Marigold (CVPR '24)[1]	Diffusion Fine-tuning	Synthetic (Hypersim/etc.)	Yes	**Medium**. Excellent dense prediction,

					but fails on glass due to additive/Lambertian assumptions.
Intrinsic Decomp.	Intrinsic Image Diffusion (CVPR '24)[5]	Probabilistic Diffusion	Synthetic Indoor	Yes	**Medium**. Generative sampling mitigates ambiguity, but lacks transmission/refraction priors.
Intrinsic Decomp.	Ouroboros (ICCV '25)[16]	Cycle-Consistent Diffusion	Unannotated + Synthetic	Yes	**High**. Cycle-consistency between inverse and forward rendering bypasses the need for ground truth.
Intrinsic Decomp.	RGB↔X (SIGGRAPH '24)[12]	Bidirectional Diffusion	Heterogeneous Datasets	Yes	**Medium**. Excellent at hallucinating missing PBR maps, but strictly constrained to opaque surfaces.
Intrinsic	IntrinsicAnything (ECCV	Diffusion Prior	3D Object	Yes	**Medium**. Excellent

Decomp.	'24)[9]	Guidance	Data		for shading/albedo decoupling, but fails on high-frequency transmissive backgrounds.
Translucent Mat.	Transparent GS (SIGGRAPH Asia '24)[54]	3D Gaussian Splatting	Multi-view images	Yes	**Low**. Highly accurate for background distortion, but completely violates the single-image constraint.
Translucent Mat.	DIAMOND-SSS (CVPR '26)[47]	Diffusion Augmented 3DGS	Sparse multi-view	Unclear	**Medium**. Demonstrates that diffusion can successfully hallucinate complex volumetric scattering.
Translucent Mat.	SeeClear (Arxiv '26)[60]	Generative Opacification	SeeClear-396k Synthetic	Yes	**High**. Algorithmically isolating transparent regions is highly

					analogous to extracting the T map.
Translucent Mat.	Flatbed SVBSDF (C&G '25)[48]	Neural Intrinsic Decomp	Flatbed Scans	Unclear	**Medium**. Directly estimates transmittance, but assumes strictly controlled scanner lighting.
Illumination Inv.	LightCity (ICCV '25)[72]	Benchmark / Dataset	Synthetic Urban	Yes	**Low**. Proves the need for consistency losses, but environments are entirely opaque.
Illumination Inv.	OLATverse (CVPR '26)[75]	Benchmark / Dataset	Real OLAT capture	Yes	**Medium**. Represents the gold standard for evaluating capture-invariant material estimation.
Test-Time Opt.	Spectral DPS (SPIE '26)[82]	Diffusion Posterior Sample	Sparse CT views	Unclear	**High**. Iterative measureme

					nt consistency can mathematically enforce $I = T \cdot$ at inference.
Test-Time Opt.	Anchored Discrete PS (Arxiv '25)[84]	Discrete Diffusion PS	Text/Image Data	Yes	**Medium**. Faster inference for hallucinating high-frequency discrete background details.
Uncertainty	DRUE (CVPR '25/'26)[86]	Difference Reconstruction	OOD Datasets	Unclear	**High**. Necessary to quantify epistemic uncertainty in highly ambiguous $T \cdot$ regions.
Datasets	3DReflecNet (CVPR '26)[90]	Benchmark / Dataset	Hybrid Synthetic/Real	Yes	**High**. Massive corpus for pre-training networks on refractive and transmissive optical

					physics.

Explicit Gaps in the Literature

Based on this exhaustive review, the hypothesis regarding the unaddressed nature of this specific problem is conclusively **confirmed**: the problem of *transmitted-layer-dominant multiplicative separation from a single uncontrolled image to yield physical material output* remains a genuine, unaddressed gap in the current computer vision and graphics literature. The literature exhibits three distinct islands of research that have not yet intersected to solve this problem:

1. **Reflection Removal** is structurally similar (layer separation) but predominantly formulated as an additive problem ($I = T + R$). While researchers acknowledge the deep complexity of reflection, they universally treat the transmitted layer as the clear "ground truth" to be recovered, completely ignoring the internal material properties (absorption, scattering, surface relief) of the transmissive medium itself[23].
2. **Intrinsic Image Decomposition** utilizes powerful generative diffusion priors to extract physical material maps (albedo, roughness, metallicity), but rigidly adheres to opaque Lambertian or Microfacet BRDF formulations. Consequently, models like Marigold[1] and Ouroboros[16] fail catastrophically when a background is visible *through* an object, as their architectures cannot separate the structural gradients of a background from the surface albedo of the foreground object.
3. **Transparent and Translucent Reconstruction** accurately models refraction, transmittance, and subsurface scattering, but heavily relies on multi-view setups (e.g., 3D Gaussian Splatting)[54], controlled illumination (flatbed scanners or OLAT light stages)[48], or simplifies the problem to depth estimation rather than material extraction[60].

The specific missing link: There is no existing framework that transforms a single in-the-wild image into logarithmic space ($\log I = \log T + \log B$) and utilizes dual-stream generative diffusion priors—one stream conditioned on natural background statistics and the other conditioned on transmissive material statistics—to perform an additive separation in log-space, subsequently converting the results back into capture-invariant physically-based rendering maps ($T(x)$ and $\sigma_s(x)$).

Foundational Reading: "Top 5 Papers"

If a research team is to build a novel architecture to solve this unaddressed multiplicative layer separation problem, the following five papers represent the critical foundation required to bridge the identified gaps:

Priority	Paper / Research Work	Rationale for Mandatory

		Reading
1	**Removing Reflections from RAW Photos** (CVPR 2025)[35]	Proves that operating in a linear, scene-referred color space (RAW) is mandatory for accurately separating mixed light fields. Attempting multiplicative layer separation on gamma-encoded, tone-mapped RGB images will fundamentally fail mathematically.
2	**WindowSeat: Reflection Removal through Efficient Adaptation of Diffusion Transformers** (NTIRE 2026)[27]	Provides the exact blueprint for generating the required training data. To succeed, researchers must build a Blender PBR pipeline simulating stained glass against random backgrounds; WindowSeat demonstrates how to efficiently train a DiT on this synthetic data using LoRA.
3	**Flash-Split: 2D Reflection Removal with Flash Cues and Latent Diffusion Separation** (CVPR 2025)[23]	The dual-branch latent diffusion architecture represents the optimal structural template for layer separation. A team can adapt this dual-stream latent approach to output the glass material maps on one branch and the background hallucination on the other.
4	**Ouroboros: Single-step Diffusion Models for Cycle-consistent Forward**	Because capturing real-world ground truth

	and Inverse Rendering (ICCV 2025)[16]	data for $T \cdot$ separation is physically impossible, cycle-consistency is required. Ouroboros demonstrates how to enforce that predicted intrinsic maps, when fed through a forward renderer, recreate the input image, bridging the synthetic-to-real gap.
5	**SeeClear: Reliable Transparent Object Depth Estimation via Generative Opacification** (Arxiv 2026)[60]	Pioneers the concept of using generative diffusion to explicitly alter the transmissive properties of an object in a 2D image. Understanding how SeeClear algorithmically "opacifies" a transparent surface will directly guide the approach to disentangling background structure from intrinsic glass color.

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